PRASAD SADANAND MAHABARE

Embedded Firmware Engineer | Firmware Testing & Validation

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PROFESSIONAL SUMMARY

Hands-on **embedded firmware development and validation** experience across **STM32, ESP32 and PIC18**, with **Embedded C, FreeRTOS**, peripheral interfaces and hardware simulation. Professional experience also includes **firmware deployment automation across 800+ network devices.**

TECHNICAL SKILLS

- **Programming:** Embedded C, C++, Python(academic), Bash
- **Embedded:** STM32F446RE, ESP32-WROOM, PIC18F4520, FreeRTOS, GPIO, ADC, Timers, Watchdog
- **Interfaces & Networking:** UART, SPI, I2C, MQTT, Ethernet, TCP/IP
- **Testing & Debugging:** Peripheral testing, serial debugging, Proteus simulation, UART diagnostics, fault troubleshooting
- **Tools:** Git, GitHub, STM32CubeIDE, MPLAB X, Arduino IDE, VS Code

EMBEDDED PROJECTS

PIC18F4520 UART Command & Peripheral Management System

- Developed an **interrupt-driven UART command interface** for controlling and **validating PIC18F4520 peripherals**.
- Implemented **serial parsing** and **interrupt handling** for reliable command processing and low-latency communication.
- **Validated peripheral behavior** in **Proteus** and debugged communication issues using **serial trace analysis**.

STM32 FreeRTOS System Health Monitor

- Developed a multitasking **STM32 Cortex-M4** application using **FreeRTOS** for scheduled monitoring and runtime diagnostics.
- Implemented **UART diagnostics**, **ADC sampling**, **heartbeat monitoring**, GPIO control, and timer-based functions.
- **Validated concurrent firmware** behavior and peripheral configuration using STM32CubeIDE.

ESPForge - OTA Firmware Management Platform

- Developed **ESP32 firmware** for **remote OTA upgrades**, **watchdog-based recovery**, and live device telemetry.
- Integrated **FreeRTOS** task orchestration with **MQTT** messaging to coordinate firmware updates and device monitoring.

PROFESSIONAL EXPERIENCE

Software Developer, Fuge Technologies **May 2024 - Sep 2025**

- Developed and maintained REST APIs in Java Spring Boot supporting **10+ automated infrastructure workflows**.
- Deployed applications using AWS EC2 and Docker, improving deployment consistency and reducing manual setup effort by **40%.**

Network Engineer, Purdue University **Jan 2023 – Dec 2023**

- Configured and maintained enterprise network infrastructure including Cisco Catalyst switches, firewalls, DNS, and load balancing systems.
- Automated **firmware deployment workflows for 800+ network devices**, reducing manual upgrade effort and improving **deployment consistency**.
- Applied security patches, hotfixes, and system updates across cloud infrastructure with a focus on operational reliability.

EDUCATION

P.G. Diploma in Embedded Systems Design **Sep 2025 - Aug 2026**

BICARD Institute, Pune

Master of Science in Computer Science **Jan 2022 - Dec 2023**

Purdue University, Fort Wayne, Indiana

Bachelor of Engineering Computer Engineering **Aug 2016 - Aug 2019**

Pune University, Pune, India